

## 第五次作业答案

### 解答题

(1)

$$\begin{aligned}
 & \int \frac{1}{\sin x + \cos x + 1} dx \\
 \stackrel{t=\tan \frac{x}{2}}{=} & \int \frac{1}{\frac{2t}{t^2+1} + \frac{1-t^2}{t^2+1} + 1} \frac{2 dt}{1+t^2} \\
 = & \int \frac{2 dt}{2t+1-t^2+1+t^2} \\
 = & \int \frac{dt}{t+1} \\
 = & \ln|t+1| + C \\
 = & \ln\left|\tan \frac{x}{2} + 1\right| + C
 \end{aligned}$$

(2)

$$\begin{aligned}
 & \int \frac{1}{\cos x + 1} dx \\
 \stackrel{t=\tan \frac{x}{2}}{=} & \int \frac{1}{\frac{1-t^2}{t^2+1} + 1} \frac{2 dt}{1+t^2} \\
 = & \int 1 dt \\
 = & t + C \\
 = & \tan \frac{x}{2} + C
 \end{aligned}$$

(3)

$$\begin{aligned}
 & \int \frac{2 \sin^3 x}{\cos 2x} dx \\
 = & \int \frac{2(1-\cos^2 x)}{2\cos^2 x - 1} (\sin x dx) \\
 = & \int \frac{2(1-\cos^2 x)}{2\cos^2 x - 1} (-d(\cos x)) \\
 \stackrel{t=\cos x}{=} & \int \frac{2(1-t^2)}{2t^2-1} (-dt) \\
 = & \int \left(1 + \frac{1}{2(1-\sqrt{2}t)} + \frac{1}{2(1+\sqrt{2}t)}\right) dt \\
 = & t + \frac{\sqrt{2}}{4} \ln \left| \frac{1+\sqrt{2}t}{1-\sqrt{2}t} \right| + C \\
 = & \cos x + \frac{\sqrt{2}}{4} \ln \left| \frac{1+\sqrt{2}\cos x}{1-\sqrt{2}\cos x} \right| + C
 \end{aligned}$$

(4)

$$\begin{aligned}
 & \int \tan^8 x dx \\
 \stackrel{t=\tan x}{=} & \int \frac{t^8 dt}{1+t^2} \\
 = & \int (t^6 - t^4 + t^2 - 1 + \frac{1}{1+t^2}) dt \\
 = & \frac{t^7}{7} - \frac{t^5}{5} + \frac{t^3}{3} - t + \arctan t + C \\
 = & \frac{\tan^7 x}{7} - \frac{\tan^5 x}{5} + \frac{\tan^3 x}{3} - \tan x + x + C
 \end{aligned}$$

(5)

$$\begin{aligned}
 & \int \frac{x-1}{x^2+6x+8} dx \\
 = & \int \frac{x-1}{(x+2)(x+4)} dx \\
 = & \int \left(-\frac{3}{2} \frac{1}{x+2} + \frac{5}{2} \frac{1}{x+4}\right) dx \\
 = & -\frac{3}{2} \ln|x+2| + \frac{5}{2} \ln|x+4| + C
 \end{aligned}$$

(6)

$$\begin{aligned}
 & \int \frac{2x^2-5}{x^4-5x^2+6} dx \\
 = & \int \frac{2x^2-5}{(x^2-2)(x^2-3)} dx \\
 = & \int \left(\frac{1}{x^2-2} + \frac{1}{x^2-3}\right) dx \\
 = & -\frac{\sqrt{2}}{4} \ln \left| \frac{\sqrt{2}+x}{\sqrt{2}-x} \right| - \frac{\sqrt{3}}{6} \ln \left| \frac{\sqrt{3}+x}{\sqrt{3}-x} \right| + C
 \end{aligned}$$

(7)

$$\begin{aligned}
 & \int \frac{x^2}{1-x^4} dx \\
 = & \int \frac{1}{2} \left( \frac{1}{1-x^2} - \frac{1}{1+x^2} \right) dx \\
 = & \frac{1}{4} \ln \left| \frac{1+x}{1-x} \right| - \frac{1}{2} \arctan x + C
 \end{aligned}$$

(8)

$$\begin{aligned}
& \int \frac{x^3}{x^4 + x^2 - 2} dx \\
&= \int \frac{x^2}{x^4 + x^2 - 2} \frac{d(x^2)}{2} \\
&\stackrel{t=x^2}{=} \int \frac{t}{t^2 + t - 2} \frac{dt}{2} \\
&= \int \left( \frac{1}{t-1} + \frac{2}{t+2} \right) \frac{dt}{6} \\
&= \frac{1}{6} \ln |t-1| + \frac{1}{3} \ln |t+2| + C \\
&= \frac{1}{6} \ln |x^2-1| + \frac{1}{3} \ln |x^2+2| + C
\end{aligned}$$

(9)

$$\begin{aligned}
& \int x^3 \ln x dx \\
&= \int \ln x \frac{d(x^4)}{4} \\
&= \frac{1}{4} x^4 \ln x - \int \frac{x^4}{4} d(\ln x) \\
&= \frac{1}{4} x^4 \ln x - \int \frac{x^3}{4} dx \\
&= \frac{1}{4} x^4 \ln x - \frac{1}{16} x^4 + C
\end{aligned}$$

(10)

$$\begin{aligned}
& \int \arctan x dx \\
&= x \arctan x - \int x d(\arctan x) \\
&= x \arctan x - \int \frac{x}{1+x^2} dx \\
&= x \arctan x - \frac{1}{2} \ln(1+x^2) + C
\end{aligned}$$

(11)

$$\begin{aligned}
& \int x \arctan x dx \\
&= \int \arctan x \frac{d(x^2)}{2} \\
&= \frac{1}{2} x^2 \arctan x - \int \frac{x^2}{2} d(\arctan x) \\
&= \frac{1}{2} x^2 \arctan x - \int \frac{x^2}{2(1+x^2)} dx \\
&= \frac{1}{2} x^2 \arctan x - \int \frac{1}{2} \left( 1 - \frac{1}{1+x^2} \right) dx \\
&= \frac{1}{2} x^2 \arctan x - \frac{1}{2} x + \frac{1}{2} \arctan x + C
\end{aligned}$$

(12)

$$\begin{aligned}
& \int \arcsin x dx \\
&= x \arcsin x - \int x d(\arcsin x) \\
&= x \arcsin x - \int \frac{x}{\sqrt{1-x^2}} dx \\
&= x \arcsin x + \sqrt{1-x^2} + C
\end{aligned}$$

(13)

$$\begin{aligned}
& \int x^2 e^{2x} dx \\
&= \int \frac{x^2}{2} d(e^{2x}) \\
&= \frac{1}{2} x^2 e^{2x} - \int e^{2x} d\left(\frac{x^2}{2}\right) \\
&= \frac{1}{2} x^2 e^{2x} - \int e^{2x} x dx \\
&= \frac{1}{2} x^2 e^{2x} - \int \frac{x}{2} d(e^{2x}) \\
&= \frac{1}{2} x^2 e^{2x} - \frac{1}{2} x e^{2x} + \int e^{2x} d\left(\frac{x}{2}\right) \\
&= \frac{1}{2} x^2 e^{2x} - \frac{1}{2} x e^{2x} + \frac{1}{4} e^{2x} \\
&= \frac{1}{4} e^{2x} (2x^2 - 2x + 1)
\end{aligned}$$

(14)

$$\begin{aligned}
I &= \int e^{2x} \sin 3x dx \\
&= \int \sin 3x d\left(\frac{e^{2x}}{2}\right) \\
&= \frac{1}{2} e^{2x} \sin 3x - \int \frac{e^{2x}}{2} d(\sin 3x) \\
&= \frac{1}{2} e^{2x} \sin 3x - \frac{3}{2} \int e^{2x} \cos 3x dx \\
&= \frac{1}{2} e^{2x} \sin 3x - \frac{3}{2} \int \cos 3x d\left(\frac{e^{2x}}{2}\right) \\
&= \frac{1}{2} e^{2x} \sin 3x - \frac{3}{4} e^{2x} \cos 3x + \frac{3}{2} \int \frac{e^{2x}}{2} d(\cos 3x) \\
&= \frac{1}{2} e^{2x} \sin 3x - \frac{3}{4} e^{2x} \cos 3x - \frac{3}{2} \int \frac{3}{2} e^{2x} \sin 3x dx \\
&= \frac{1}{2} e^{2x} \sin 3x - \frac{3}{4} e^{2x} \cos 3x - \frac{9}{4} I \\
\therefore I &= \frac{2}{13} e^{2x} \sin 3x - \frac{3}{13} e^{2x} \cos 3x
\end{aligned}$$

(15)

$$\begin{aligned} I_n &= \int \sin^n x \, dx \\ &= - \int \sin^{n-1} x \, d(\cos x) \\ &= - \sin^{n-1} x \cos x + \int \cos x \, d(\sin^{n-1} x) \\ &= - \sin^{n-1} x \cos x + \int \cos x (n-1) \sin^{n-2} x \cos x \, dx \\ &= - \sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x (1 - \sin^2 x) \, dx \\ &= - \sin^{n-1} x \cos x + (n-1)(I_{n-2} - I_n) \end{aligned}$$

$$\therefore I_n = -\frac{1}{n} \sin^{n-1} x \cos x + \frac{n-1}{n} I_{n-2} \quad (n \geq 2)$$

又已知  $I_0 = \int \sin^0 x \, dx = \int 1 \, dx = x + C$ ,  $I_1 = \int \sin^1 x \, dx = -\cos x + C$ , 由此可以递推求出  $I_n$ .

## 复习题

以下题目不要求完成, 可作为复习习题

1.  $\int \frac{1}{x \ln x} \, dx = \ln(\ln x) + C$

2.  $\lim_{x \rightarrow 0} \frac{2x^2}{1 - \cos 3x} = \frac{4}{9}$

3.  $\lim_{h \rightarrow 0} \frac{\sin(x+h)^2 - \sin x^2}{h} = (\sin x^2)' = 2x \cos x^2$

4. 排序:  $2^{\frac{1}{2}}, 3^{\frac{1}{3}}, 6^{\frac{1}{6}}$

(way 1): 考虑对所有元素的六次方进行排序:  $2^3, 3^2, 6$ .

显然有  $6 < 2^3 = 8 < 3^2 = 9$ .

由于  $y = \sqrt[6]{x}$  在  $(0, +\infty)$  上单调递增, 所以  $6^{\frac{1}{6}} < 2^{\frac{1}{2}} < 3^{\frac{1}{3}}$ .

(way 2): 考虑对取对数之后的三个数排序:  $\frac{\ln 2}{2}, \frac{\ln 3}{3}, \frac{\ln 6}{6}$ .

定义函数  $f(x) = \frac{\ln x}{x}$ . 则需要对  $f(2), f(3), f(6)$  进行排序.

容易证明  $f(x)$  在  $(0, e)$  上单调递增, 在  $(e, +\infty)$  上单调递减.

由于  $2 < e < 3$ , 所以可以用单调性得出  $f(6) < f(3)$ ,

但是单调性无法判断出  $f(6)$  与  $f(2)$  谁大, 以及  $f(3)$  与  $f(2)$  谁大. 因此还需要借助 way 1 的方法判断.

**Remark:** 当需要排序的数字更大, 如  $100^{\frac{1}{100}}$  时, way 1 基本不可行, 只能用 way 2 的方法. 所以最好同时掌握这两种方法.

5.  $\int_{-1}^1 x^2 (\sin x + 1) \, dx = \frac{2}{3}$

6.  $\int_{-1}^1 \frac{1}{1 + e^x} \, dx = 1$

7.  $f(x) = \frac{x-1}{e^x}$ , 则  $f^{(2025)}(x) = \frac{2026-x}{e^x}$

8.  $\int \frac{e^{ax}}{1 + e^{ax}} \, dx = \frac{1}{a} \ln(1 + e^{ax}) + C$

9.  $\int \frac{1}{1 + e^{ax}} \, dx = x - \frac{1}{a} \ln(1 + e^{ax}) + C$

$$10. \int \frac{e^{ax} - e^{bx}}{(1 + e^{ax})(1 + e^{bx})} dx = \int \frac{(e^{ax} + 1) - (e^{bx} + 1)}{(1 + e^{ax})(1 + e^{bx})} dx = \int \frac{1}{1 + e^{bx}} dx - \int \frac{1}{1 + e^{ax}} dx = \frac{1}{b} \ln(1 + e^{bx}) - \frac{1}{a} \ln(1 + e^{ax}) + C$$

$$11. \text{ 已知 } f(x) = \begin{cases} -x^2 + 4x - 2 & x \geq 1 \\ -2x^2 + 3 & x < 1 \end{cases}, \text{ 求 } f(x) \text{ 的极值和最值.}$$

$f(x)$  的极大值:  $f(2) = 2, f(0) = 3$ ; 极小值:  $f(1) = 1$ . 最大值  $f(0) = 3$ , 没有最小值.

$$12. a \text{ 取什么值的时候, 积分 } \int_a^{a+2} (\ln x - x) dx \text{ 有最大值?}$$

考虑函数  $f(a) = \int_a^{a+2} (\ln x - x) dx, (a > 0)$ . 则  $f'(a) = \ln(a+2) - (a+2) - (\ln a - a) = \ln(1 + \frac{2}{a}) - 2$ . 因此,

$a = \frac{2}{e^2 - 1}$  时,  $f(a)$  取到最大值.